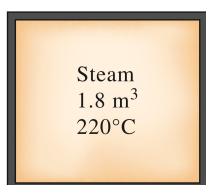
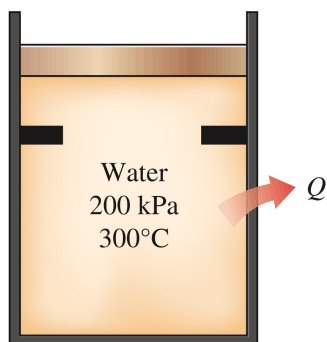


แนวข้อสอบ heat รุ่น 32
ปีผมออกตามนี้นะ อ่านเป็นแนวๆ
ปีอื่นอาจจะออกข้ออื่นก็ได้

3–28 A 1.8-m^3 rigid tank contains steam at 220°C . One-third of the volume is in the liquid phase and the rest is in the vapor form. Determine (a) the pressure of the steam, (b) the quality of the saturated mixture, and (c) the density of the mixture.

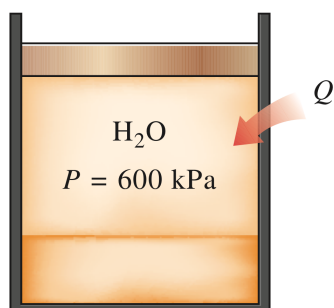


3–43 Water initially at 200 kPa and 300°C is contained in a piston-cylinder device fitted with stops. The water is allowed to cool at constant pressure until it exists as a saturated vapor and the piston rests on the stops. Then the water continues to cool until the pressure is 100 kPa . On the T - v diagrams sketch, with respect to the saturation lines, the process curves passing through both the initial, intermediate, and final states of the water. Label the T , P and v values for end states on the process curves. Find the overall change in internal energy between the initial and final states per unit mass of water.

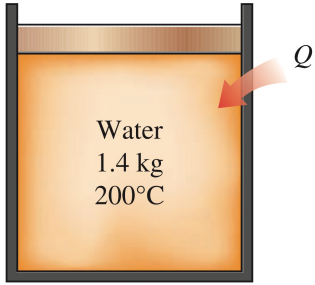


3–51  A piston-cylinder device contains 0.005 m^3 of liquid water and 0.9 m^3 of water vapor in equilibrium at 600 kPa . Heat is transferred at constant pressure until the temperature reaches 200°C .

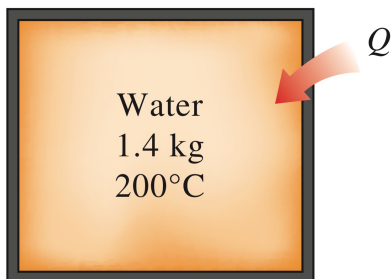
- What is the initial temperature of the water?
- Determine the total mass of the water.
- Calculate the final volume.
- Show the process on a P - v diagram with respect to saturation lines.



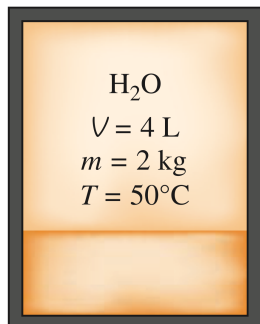
3–59 A piston-cylinder device initially contains 1.4-kg saturated liquid water at 200°C. Now heat is transferred to the water until the volume quadruples and the cylinder contains saturated vapor only. Determine (a) the volume of the tank, (b) the final temperature and pressure, and (c) the internal energy change of the water.



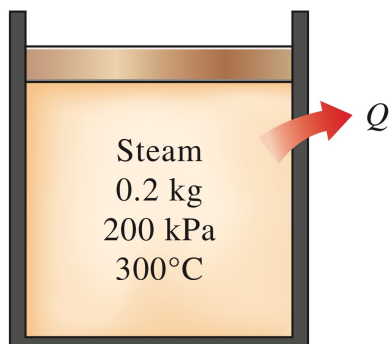
3–63 A rigid tank initially contains 1.4-kg saturated liquid water at 200°C. At this state, 25 percent of the volume is occupied by water and the rest by air. Now heat is supplied to the water until the tank contains saturated vapor only. Determine (a) the volume of the tank, (b) the final temperature and pressure, and (c) the internal energy change of the water.



3–118 A 4-L rigid tank contains 2 kg of saturated liquid–vapor mixture of water at 50°C. The water is now slowly heated until it exists in a single phase. At the final state, will the water be in the liquid phase or the vapor phase? What would your answer be if the volume of the tank were 400 L instead of 4 L?

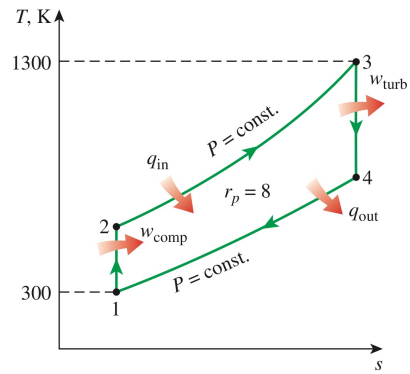


3–120 A piston-cylinder device initially contains 0.2 kg of steam at 200 kPa and 300°C. Now, the steam is cooled at constant pressure until it is at 150°C. Determine the volume change of the cylinder during this process using the compressibility factor and compare the result to the actual value.

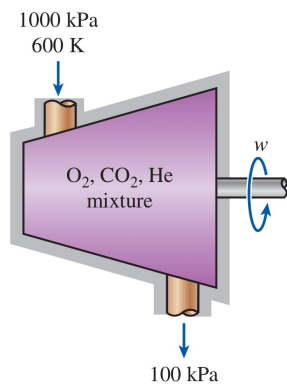


EXAMPLE 9–5 The Simple Ideal Brayton Cycle

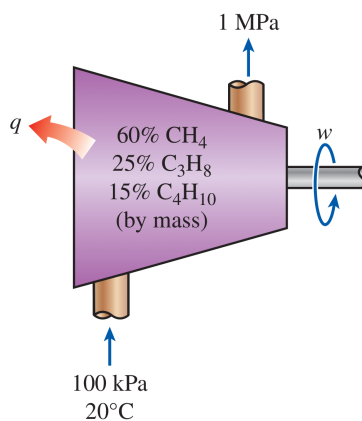
A gas-turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Utilizing the air-standard assumptions, determine (a) the gas temperature at the exits of the compressor and the turbine, (b) the back work ratio, and (c) the thermal efficiency.



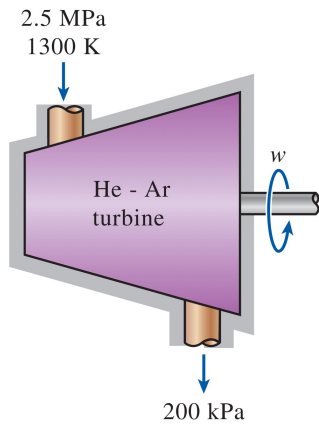
A mixture of oxygen (O_2), carbon dioxide (CO_2), and helium (He) gases with mass fractions of 0.0625, 0.625, and 0.3125, respectively, enter an adiabatic turbine at 1000 kPa and 600 K steadily and expand to 100 kPa pressure (Fig. 13–14). The isentropic efficiency of the turbine is 90 percent. For gas components assuming constant specific heats at room temperature, determine (a) the work output per unit mass of mixture and (b) the exergy destruction and the second-law efficiency of the turbine. Take the environment temperature to be $T_0 = 25^\circ\text{C}$.



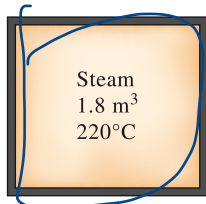
13–59 A mixture of hydrocarbon gases is composed of 60 percent methane, 25 percent propane, and 15 percent butane by weight. This mixture is compressed from 100 kPa and 20°C to 1000 kPa in a reversible, isothermal, steady-flow compressor. Calculate the work and heat transfer for this compression per unit mass of the mixture.



13–60 An equimolar mixture of helium and argon gases is to be used as the working fluid in a closed-loop gas-turbine cycle. The mixture enters the turbine at 2.5 MPa and 1300 K and expands isentropically to a pressure of 200 kPa. Determine the work output of the turbine per unit mass of the mixture.



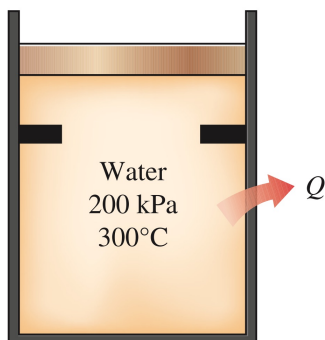
3-28 A 1.8-m³ rigid tank contains steam at 220°C. One-third of the volume is in the liquid phase and the rest is in the vapor form. Determine (a) the pressure of the steam, (b) the quality of the saturated mixture, and (c) the density of the mixture.



known
Sat. Mix
 $V = 1.8 \text{ m}^3$
 $T = 220^\circ\text{C}$
 $V_f = \frac{1}{3}V$
 $V_g = \frac{2}{3}V$

find
 $P = 2114.6 \text{ kPa}$
 $V_f = 0.6 \text{ m}^3$
 $V_g = 1.2 \text{ m}^3$
 $m_f = \frac{V_f}{v_f} = \frac{0.6}{0.00119} = 504.2$
 $m_g = \frac{V_g}{v_g} = \frac{1.2}{0.016042} = 74.8$
 $x = \frac{m_g}{m_f} = \frac{74.8}{504.2} = 0.148$
 $\rho = \frac{m}{V} = \frac{504.2 + 74.8}{1.8} = 319.16$

3-43 Water initially at 200 kPa and 300°C is contained in a piston-cylinder device fitted with stops. The water is allowed to cool at constant pressure until it exists as a saturated vapor and the piston rests on the stops. Then the water continues to cool until the pressure is 100 kPa. On the T - v diagrams sketch, with respect to the saturation lines, the process curves passing through both the initial, intermediate, and final states of the water. Label the T , P and v values for end states on the process curves. Find the overall change in internal energy between the initial and final states per unit mass of water.

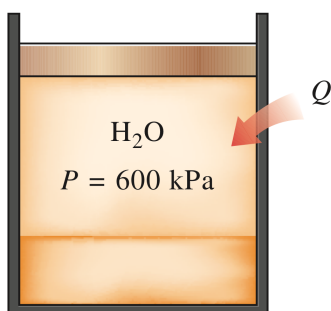


known
State 1: $P = 200 \text{ kPa}$
 $T = 300^\circ\text{C}$
State 2: $P = 200 \text{ kPa}$ Sat. Vap
State 3: $P = 100 \text{ kPa}$

find
Superheated
 $u_1 = 2808.8$
 $v_1 = 1.31663$
 $T_2 = 150.06^\circ\text{C}$, $u_2 = u_g = 2582.1$
 $v_2 = v_g = 0.35206$
Sat. Mix
 $v_3 = v_f = 0.001043$ Sat. Mix
 $x = \frac{v - v_f}{v_g - v_f} = \frac{0.35206 - 0.001043}{1.04041 - 0.001043} = 0.343$
 $u_3 = u_f + x u_{fg}$
 $= 167.8 + 0.343(2019.23)$
 $= 854.52$
 $\Delta u = u_3 - u_1 = 854.52 - 2808.8 = -1954.28 \text{ kJ/kg}$

3-51 A piston-cylinder device contains 0.005 m³ of liquid water and 0.9 m³ of water vapor in equilibrium at 600 kPa. Heat is transferred at constant pressure until the temperature reaches 200°C.

- What is the initial temperature of the water?
- Determine the total mass of the water.
- Calculate the final volume.
- Show the process on a P - v diagram with respect to saturation lines.



known
State 1: $V_f = 0.005$
 $V_g = 0.9$
 $P = 600 \text{ kPa}$

find
Sat. Mix
 $T_1 = 158.83^\circ\text{C}$
 $v_f = 0.001101$
 $v_g = 0.31560$
 $m_f = \frac{V_f}{v_f} = \frac{0.005}{0.001101} = 4.54$
 $m_g = \frac{V_g}{v_g} = \frac{0.9}{0.31560} = 2.85$
 $m = m_f + m_g = 7.39$
Superheated
 $T_2 = 200^\circ\text{C}$
 $v_2 = 0.31560$
 $V_2 = m v_2 = 7.39 \times 0.31560 = 2.33$

3-59 A piston-cylinder device initially contains 1.4-kg saturated liquid water at 200°C. Now heat is transferred to the water until the volume quadruples and the cylinder contains saturated vapor only. Determine (a) the volume of the tank, (b) the final temperature and pressure, and (c) the internal energy change of the water.



known	find
State 1; Sat. liq. $x=0$ $m_1 = 1.4 \text{ kg}$ $T_1 = 200^\circ\text{C}$	$v_f = 0.001153$ $v_f = m f = 1.4 (0.001153)$ $= 0.001615$ $u_1 = 160.45$
State 2; Sat. Vap. $x=1$ $v = 4v_1$ $m_2 = 1.4 \text{ kg}$	$v_g = x(0.001153) = 0.001615$ $v_g = \frac{v_g}{m} = \frac{0.001615}{1.4} = 0.001153$ $T_2 = 170.4$ $P_2 = 1012.4$ $u_g = 2142$
$\Delta u = m(u_2 - u_1) = (2142 - 160.45)(1.4) = 1111$	

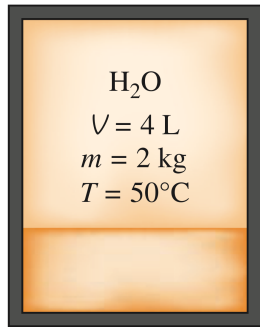
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3-63 A rigid tank initially contains 1.4-kg saturated liquid water at 200°C. At this state, 25 percent of the volume is occupied by water and the rest by air. Now heat is supplied to the water until the tank contains saturated vapor only. Determine (a) the volume of the tank, (b) the final temperature and pressure, and (c) the internal energy change of the water.



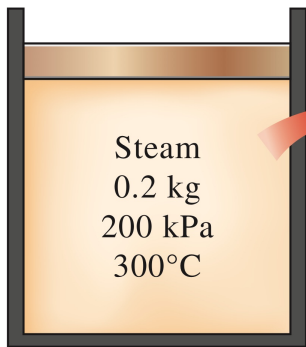
State 1; Sat. liq. $x=0$ $m_1 = 1.4 \text{ kg}$ $T_1 = 200^\circ\text{C}$ $v_1 = \frac{1}{4} v_t$	$v_f = 0.001153$ $v_f = m f = 1.4 (0.001153)$ $= 0.001615$ $v_t = 4v_f = 0.00646$
State 2; Sat. Vap. $x=1$ $m_2 = 1.4 \text{ kg}$ $v_2 = 0.001153$	

3-118 A 4-L rigid tank contains 2 kg of saturated liquid-vapor mixture of water at 50°C. The water is now slowly heated until it exists in a single phase. At the final state, will the water be in the liquid phase or the vapor phase? What would your answer be if the volume of the tank were 400 L instead of 4 L?



	known	find
State 1;	Sat Mix $m: 2 \text{ kg}$ $T: 50^\circ\text{C}$ $V: 0.004$	$v: \frac{V}{m} = \frac{0.004}{2} = 0.002$ $v < v_{cr} : \text{liquid phase}$
State 2; ?		$v: \frac{V}{m} = \frac{0.4}{2} = 0.2$ $v > v_{cr} : \text{Vapor phase}$

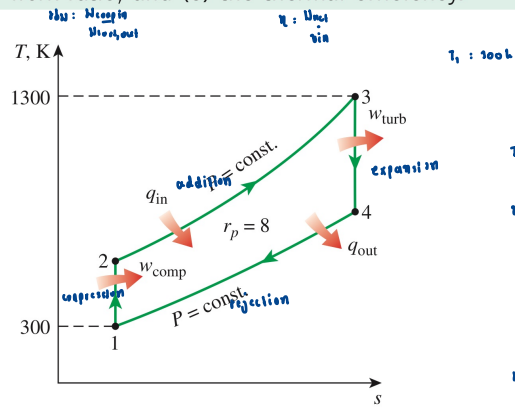
3-120 A piston-cylinder device initially contains 0.2 kg of steam at 200 kPa and 300°C. Now, the steam is cooled at constant pressure until it is at 150°C. Determine the volume change of the cylinder during this process using the compressibility factor and compare the result to the actual value.



	known	find
State 1;	$m: 0.2$ $P: 200 \text{ kPa}$ $T: 300$	superheated $v: 1.51553$ $v \cdot m = 0.303106$ $: 0.3032$
State 2;	$P: 200 \text{ kPa}$ $T: 150$	superheated $v: 0.454956$ $v \cdot m = 0.0909912$ $: 0.091$
		$\Delta V = v_2 - v_1 = 0.091 - 0.3032$ $: -0.2122$

EXAMPLE 9-5 The Simple Ideal Brayton Cycle

A gas-turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Utilizing the air-standard assumptions, determine (a) the gas temperature at the exits of the compressor and the turbine, (b) the back work ratio, and (c) the thermal efficiency.



$$T_1 : 300 \text{ K} \rightarrow h : 300.19 \text{ kJ/kg}$$

$$P_2 : 1.1010 \text{ bar}$$

$$T : P_2 \rightarrow P_2 : 1.1010 \rightarrow 11.019 \text{ bar}$$

$$T_2 : 551.16 \text{ K}$$

$$h_2 : 508.90 \text{ kJ/kg}$$

$$T_1 : 1300 \text{ K} \rightarrow h : 1335.43 \text{ kJ/kg}$$

$$P_2 : 1.1010 \text{ bar}$$

$$T : P_2 \rightarrow P_2 : 1.1010 \rightarrow 11.019 \text{ bar}$$

$$T_2 : 551.16 \text{ K}$$

$$h_2 : 508.90 \text{ kJ/kg}$$

$$\eta_{th} = \frac{W_{net, out}}{W_{turb, in}} = \frac{h_3 - h_4}{h_3 - h_2} = \frac{1335.43 - 719.55}{1335.43 - 508.90} = \frac{615.88}{826.53} = 0.745$$

$$W_{net} : W_{turb, out} - W_{comp, in}$$

$$: 601.54 - 205.61$$

$$: 395.93$$

$$W_{in} : h_2 - h_1 : 508.90 - 300.19$$

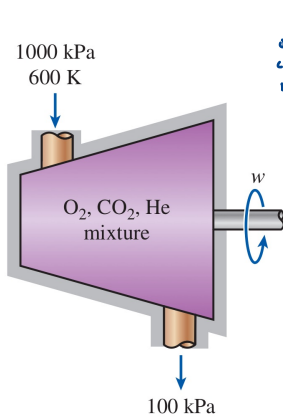
$$: 208.71$$

$$\eta : \frac{W_{net}}{W_{in}} = \frac{395.93}{524.62} = 0.755$$

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A mixture of oxygen (O_2), carbon dioxide (CO_2), and helium (He) gases with mass fractions of 0.0625, 0.625, and 0.3125, respectively, enter an adiabatic turbine at 1000 kPa and 600 K steadily and expand to 100 kPa pressure (Fig. 13–14). The isentropic efficiency of the turbine is 90 percent. For gas components assuming constant specific heats at room temperature, determine (a) the work output per unit mass of mixture and (b) the exergy destruction and the second-law efficiency of the turbine. Take the environment temperature to be $T_0 = 25^\circ\text{C}$.

$$n_i \rightarrow c_p n_i, c_{v,n} \rightarrow h \rightarrow T_{2s} \rightarrow T_2 \rightarrow W$$



$$n_i$$

$$\begin{aligned} m_{O_2} &= 0.0625 \\ m_{CO_2} &= 0.625 \\ m_{He} &= 0.3125 \end{aligned}$$

$$c_p n_i = \sum c_p n_i$$

$$= 0.0625 (0.918) + 0.625 (0.845) + 0.3125 (5.191)$$

$$= 2.204$$

$$c_{v,n} = \sum c_{v,n} i$$

$$= 0.0625 (0.659) + 0.625 (0.659) + 0.3125 (1.916)$$

$$= 1.818$$

$$k = \frac{c_p n_i}{c_{v,n}} = \frac{2.204}{1.818} = 1.21$$

$$T_{2s} = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}} = 600 \left(\frac{100}{1000} \right)^{\frac{1.21-1}{1.21}}$$

$$= 311.0 \text{ K}$$

$$\eta = \frac{T_1 - T_2}{T_1 - T_{2s}}$$

$$T_2 = T_1 - \eta (T_1 - T_{2s})$$

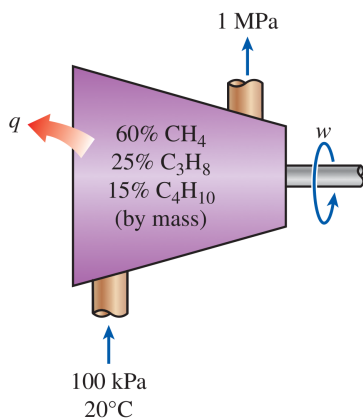
$$= 600 - 0.9 (600 - 311.0)$$

$$= 340.1 \text{ K}$$

$$W = c_p n_i (T_1 - T_2) = 2.204 (600 - 340.1)$$

$$= 575.4 \text{ kJ/kg}$$

13–59 A mixture of hydrocarbon gases is composed of 60 percent methane, 25 percent propane, and 15 percent butane by weight. This mixture is compressed from 100 kPa and 20°C to 1000 kPa in a reversible, isothermal, steady-flow compressor. Calculate the work and heat transfer for this compression per unit mass of the mixture.



$$n_i = 0.6, 0.25, 0.15$$

$$n_i = \sum n_i = \frac{1}{28.9} + \frac{1}{44.1} + \frac{1}{58.1} = 0.035 + 0.023 + 0.017 = 0.075$$

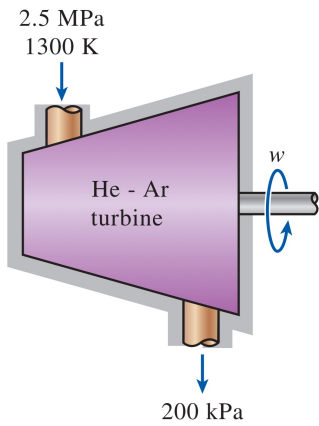
$$n_i = \frac{1}{28.9} = 0.035$$

$$W = \sum n_i \ln \left(\frac{P_2}{P_1} \right) = 0.075 (20 + 28.9) \ln \left(\frac{1000}{100} \right)$$

$$= 265.7 \text{ kJ/kg}$$

$$Q = W = 265.7 \text{ kJ/kg}$$

13–60 An equimolar mixture of helium and argon gases is to be used as the working fluid in a closed-loop gas-turbine cycle. The mixture enters the turbine at 2.5 MPa and 1300 K and expands isentropically to a pressure of 200 kPa. Determine the work output of the turbine per unit mass of the mixture.



$$\begin{aligned}
 M_m &= \sum M_i : 0.5(4.003) + 0.5(39.948) \\
 &= 22 \\
 m_{He} &= \frac{M_i}{M_m} : \frac{0.5(4.003)}{22} = 0.09 \\
 m_{Ar} &= \frac{M_i}{M_m} : \frac{0.5(39.948)}{22} = 0.909 \\
 C_{pm} &= \sum C_{pi} m_i \\
 &= 4.191(0.09) + 0.519(0.909) \\
 &= 0.4943 \\
 C_{vm} &= \sum C_{vi} m_i \\
 &= 3.115(0.09) + (0.404)(0.909) \\
 &= 0.564 \\
 k &= \frac{C_{pm}}{C_{vm}} : \frac{0.4943}{0.564} = 1.147 \\
 T_2 = T_{2s} &= T_1 \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}} : 1300 \left(\frac{200}{2500} \right)^{\frac{1.147-1}{1.147}} \\
 &= 491.913 \\
 W &= C_{pm} (T_1 - T_2) \\
 &= 0.4943 (1300 - 491.913) \\
 &= 399.16 \text{ kJ/kg}
 \end{aligned}$$

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